

FastAPI Topics Covered

1. FastAPI setup and installation

- Installing FastAPI and Uvicorn
- Creating a FastAPI application
- Running the server with Uvicorn

2. Basic API / GET request

- `@app.get()`
- Root endpoint /

3. Path Parameters

- `/student/{student_id}`
- Getting a specific student using ID

4. Query Parameters

- `/search?course=B.Pharm`
- Optional query parameters

5. POST Request

- Adding a new student
- Request body

6. Pydantic Models

- `BaseModel`
- Data validation
- `Field()`

7. PUT Request

- Updating the complete student record

8. DELETE Request

- Deleting a student

9. PATCH Request

- Partially updating a student

10. HTTPException

- Handling errors

- 404 Student not found

11. Response Models

- response_model
- Controlling API response structure

12. Swagger / API Documentation

- /docs
- Testing GET, POST, PUT, PATCH and DELETE APIs

13. Filtering

- Filter by course
- Filter by minimum marks
- Multiple filters together

14. Pagination

- page
- limit
- start
- end
- Page-by-page results

15. Student Statistics

- Total students
- Average marks
- Highest marks
- Lowest marks
- Students by course